UI-TARS: Pioneering Automated GUI Interaction with Native Agents

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Paper version: arXiv 2501.12326v1 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *The transition from agent frameworks to agent models introduces a fundamental data bottleneck.*

A native agent must fold perception, action, reasoning and memory into **one** model.

Motivation: three gaps

1. **Handcrafted frameworks.** GPT-4o plus expert prompts and modules.
2. **Text inputs.** HTML and a11y trees vary by platform.
3. **Data.** End-to-end training needs unified traces.

Evaluation setup

OSWorld · 369 tasks | AndroidWorld · 116 tasks | Qwen-2-VL 7B / 72B | ~50B training tokens | 15 / 50-step budgets | OSWorld: 3 runs averaged

- **Online** results charted here; offline step and grounding tables are kept apart.
- **Not stated:** whether OSWorld apps overlap the online-trace tasks.

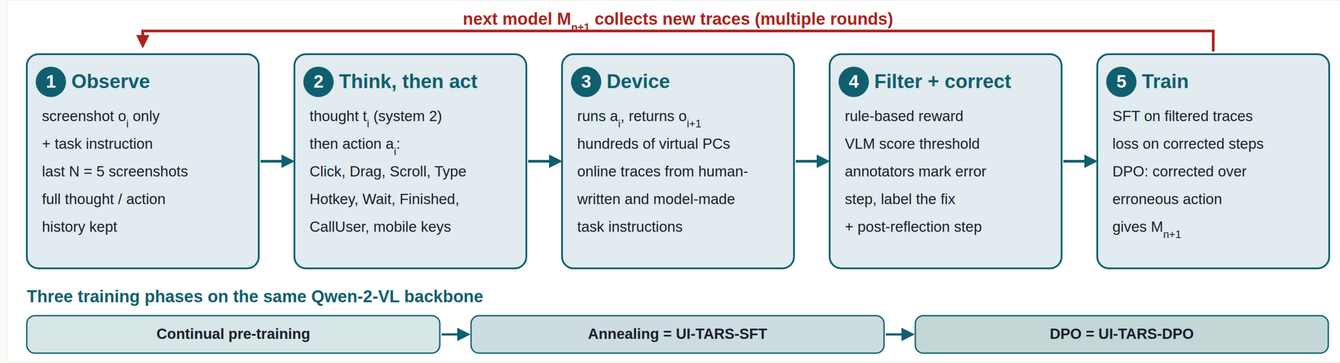
Our read (CUA Radar)

67	★★	transfer
score of 100	band	evidence level

- **Reports:** online eval; same step budgets vs Claude; 3-run mean; SFT vs DPO; ~50B tokens.
- **Not reported:** held-out environments, variance (std / CI), validation of the VLM trace filter.
- Pattern K25 (action traces + generated thoughts). Authors' numbers, not rerun.

Method: one screenshot-to-action model, trained on its own corrected traces

- **Sees / acts:** screenshots only (last N = 5); click, drag, scroll, type, hotkey out.
- **New:** GUI perception data, unified actions, a thought before each action, online traces.
- **Signal:** SFT on filtered traces and corrections, then DPO on corrected over erroneous steps.



Our redraw (not the paper's figure): the per-step loop, online trace collection and the training stages.

$$L_{\mathrm{DPO}}(\theta) = -\mathbb{E}_{\tau} \log \sigma\left(\beta \log \frac{\pi_\theta(a'_\tau|s_\tau)}{\pi_{\mathrm{SFT}}(a'_\tau|s_\tau)} - \beta \log \frac{\pi_\theta(a_\tau|s_\tau)}{\pi_{\mathrm{SFT}}(a_\tau|s_\tau)}\right)$$

DPO: prefer the corrected action a' over the erroneous a

Key results: ONLINE benchmark score

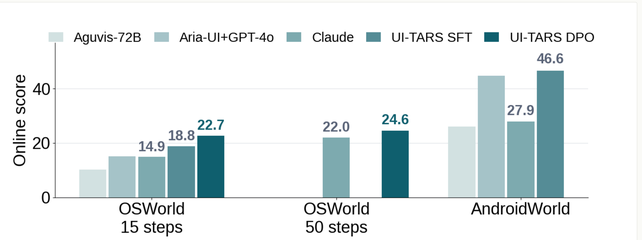


Table 9, redrawn: UI-TARS = 72B; OSWorld (screenshot-only) and AndroidWorld; blank bar = not reported.

72B-DPO 24.6 vs Claude 22.0 (50 steps)

SO WHAT → *Unlabelled OSWorld rows follow Table 9's 15-step caption.*

SFT vs DPO: OSWorld, 15 steps

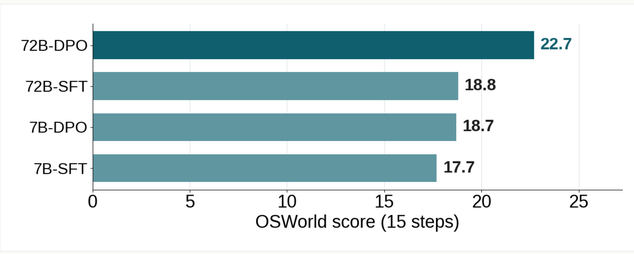


Table 9, redrawn: same model size, DPO phase on or off.

SO WHAT → *DPO raises OSWorld at both sizes: 72B 18.8 to 22.7, 7B 17.7 to 18.7. DPO is evaluated only on OSWorld; other benchmarks use the SFT model. No other component ablation is reported. Offline, in-domain: system 2 is slightly below system 1 at N = 1 and clearly ahead at N = 16 and 64 (best-of-N); on out-of-domain AndroidWorld (N = 1) system 2 is significantly better.*

Headline Numbers

24.6
OSWorld, 50 steps
72B-DPO; Claude 22.0

- **22.7** OSWorld, 15 steps
- **46.6** AndroidWorld, SFT
- **18.7** 7B-DPO, OSWorld 15

The paper's own controls

CONTROL	UI-TARS	COMPARED WITH
OSWorld, 15 steps	72B-DPO 22.7	Claude 14.9
OSWorld, 50 steps	72B-DPO 24.6	Claude 22.0
DPO vs SFT, OSWorld 15	72B 22.7 / 7B 18.7	SFT 18.8 / 17.7
AndroidWorld (OOD)	72B-SFT 46.6	Aria-UI + GPT-4o 44.8

Takeaway

A screenshot-only native agent reports 24.6 on OSWorld at 50 steps and 22.7 at 15 (Claude: 22.0 and 14.9); held-out status and variance are not reported.

~50B tokens, Qwen-2-VL | DPO helps OSWorld | AndroidWorld: SFT only | no std / CI reported

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Paper | Code